

RECRUITAI

PROJECT SPECIFICATION & IMPLEMENTATION BLUEPRINT

Event Management Platform

Detailed project brief generated from the project record.

CANDIDATE

Varun Chopra

DURATION

14 days

TECHNOLOGY STACK

Python

React

MySQL

REST API

Project Overview

Summary of the problem, purpose and expected outcome.

PROJECT SUMMARY

Web app for creating public events, user registration, and admin oversight. Implements a Python REST API, React UI, and MySQL storage.

CANDIDATE Varun Chopra	TYPE fullstack	DURATION 14 days
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Primary Objectives

- Implement JWT based authentication.
- Create CRUD for events.
- Enable users to register/unregister for events.
- Build an admin dashboard for user and event analytics.

Candidate Details

NAME

Varun Chopra

Technology Stack

Technologies, frameworks and tools expected for implementation.

Python

Technology / Tool

React

Technology / Tool

MySQL

Technology / Tool

REST API

Technology / Tool

Functional Requirements

- Users can sign up, log in, and receive a JWT.
- Authenticated users can create, edit, list, and delete their events.
- Anyone can view public events.
- Logged in users can register for an event and view their registrations.
- Admins can view all users, events, and registration counts.
- System sends email confirmation on successful registration.

Non-Functional Requirements

- All API responses under 200 ms for typical queries.
- Password stored with bcrypt hashing.
- Input validation on both client and server.
- Responsive design for desktop and mobile browsers.
- Basic logging of errors and access events.

Modules & Features

Major functional areas that make up the project.

Authentication

Handles user sign up, login, JWT issuance, and role management.

- Sign up
- Login
- Password hashing
- Role assignment

Event Management

CRUD operations for events owned by users.

- Create event
- Edit event
- Delete event
- List events

Registration

Allows users to join events and manage their registrations.

- Register
- Cancel registration
- View my registrations

Admin Dashboard

Provides overview metrics and user management tools.

- User list
- Event analytics

Implementation Roadmap

Recommended phases and milestones for completing the project.

1

Foundation

Initialize Git repo, Create Dockerfiles, Configure CI pipeline, Set up MySQL container, Scaffold Flask app, Scaffold React app

- Initialize Git repo
- Create Dockerfiles
- Configure CI pipeline
- Set up MySQL container
- Scaffold Flask app

2

Core API

Design DB schema, Implement user model, Add sign up/login routes, Create event CRUD routes, Add JWT middleware, Write basic unit tests

- Design DB schema
- Implement user model
- Add sign up/login routes
- Create event CRUD routes
- Add JWT middleware

3

Frontend MVP

Set up React router, Create auth context, Build login/signup pages, Build event list page, Integrate API calls, Add form validation

- Set up React router
- Create auth context
- Build login/signup pages
- Build event list page
- Integrate API calls

4

Advanced Features

Implement registration API, Create registration UI, Build admin dashboard components, Add email confirmation stub, Implement role based UI guards, Write integration tests

- Implement registration API
- Create registration UI
- Build admin dashboard components
- Add email confirmation stub
- Implement role based UI guards

5

Quality & Deploy

Write end to end tests with Cypress, Add error handling middleware, Configure logging, Create deployment scripts, Run performance sanity checks, Finalize documentation

- Write end to end tests with Cypress
- Add error handling middleware
- Configure logging
- Create deployment scripts
- Run performance sanity checks

API Design

Key API endpoints and expected responsibilities.

POST

/api/auth/signup

Create new user

POST

/api/auth/login

Authenticate and return JWT

GET

/api/events

List public events

POST

/api/events

Create event (auth)

PUT

/api/events/:id

Update own event

DELETE `/api/events/:id`

Delete own event

POST `/api/events/:id/register`

Register for event

GET `/api/admin/metrics`

Admin analytics

Evaluation Criteria

Criteria that can be used to assess the completed project.

- Criterion: Correctness | Description: All functional requirements work as specified | Weight: 30
- Criterion: Code Quality | Description: Clean, readable, and well structured code | Weight: 15
- Criterion: Architecture | Description: Logical separation of concerns and proper layering | Weight: 15
- Criterion: Testing | Description: Comprehensive unit, integration, and e2e tests | Weight: 15
- Criterion: Security | Description: Proper authentication, password hashing, and input validation | Weight: 15
- Criterion: Documentation | Description: Clear README, API docs, and deployment guide | Weight: 10

Implementation Note

This document is generated from the project record. AI-generated requirements should be reviewed by an administrator before implementation.